

10 > Of the SSDs and HDDs

indicators or data transfer. If the SSD is able to power up, the issue may be caused by a software misconfiguration or a setup problem. To avoid interfering with the BIOS or formatting your drive, check that everything is correctly aligned and that all cables are connected before you start.

Consider peripheral connections. Although an external device may be causing the problem, it's better to remove all peripherals that aren't essential for basic computer operations and see if this resolves the SSD issue.

Software And File System Issues

It's possible that a malfunctioning application, rather than the SSD, is responsible for errors and failures. To troubleshoot the issue, reboot your computer in Safe Mode and check if the SSD problems persist. Safe Mode operates with a minimal set of drivers and can help determine if your computer's software is at fault.

S.M.A.R.T. Failures

S.M.A.R.T is a useful technology that can perform self-diagnostics and assess the health of modern hard drives/SSDs. However, it may not be very effective in detecting sudden SSD failure. Instead, it is more commonly used to monitor the gradual decline in drive performance due to normal wear and tear. To obtain S.M.A.R.T metrics, you will need to use third-party software such as Speedfan.

Bad Blocks

Just like hard drives, SSDs can encounter an issue known as "bad sectors." In the case of SSDs, these are referred to as "bad blocks." Bad blocks can lead to various problems such as data loss, file system corruption, errors while reading files, and more, which we will delve into later in this discussion.

File Cannot Be Written Or Read

If attempting to open or save a file results in a long wait time or an error message stating "The finder can't complete the operation because some data can't be read or written," it is probable that there is file system corruption.

There are two possible scenarios when dealing with bad blocks on an SSD:

1. The system detects a bad block while writing data to the disk, which means the file was not actually written to the disk and is not damaged. This SSD problem is usually resolved automatically by the system, but if not, you can try saving the file to the cloud, restarting your computer, and then returning the data to its original location.
2. The system detects a bad block after the data has been written, so it refuses to read the file. Unfortunately, in this scenario, the data cannot be easily restored. You can attempt some methods to retrieve data from a faulty SSD, but bad blocks



often mean that the information contained in these blocks is lost permanently.

Another indication of file system corruption is receiving a message that says "Your PC needs to be repaired". Fortunately, fixing this problem is simple and doesn't require any special expertise. Windows, macOS, and Linux all have built-in tools for repairing a damaged file system. Whenever such an error occurs, the operating system will prompt you to run the appropriate tool. Simply follow the instructions provided and wait patiently until the process is complete.

Frequent Crashes

Frequent crashes are a common sign of a failing SSD, but they can also be caused by various other computer issues. However, if your computer experiences failures during boot-up or only works after several restarts, it may indicate that the SSD is nearing its end and there may not be much you can do to fix it. If you don't want to purchase a new drive immediately, be sure to back up your data and reinstall Windows as soon as possible.

Now that you've understood how to troubleshoot SSD issues, it's time to understand how you can fix issues related to HDD.

Three ways to tell if your HDD is failing

Hard drives typically fail gradually, providing enough time to retrieve data and replace them before a catastrophic failure. However, how can you tell if your hard drive is failing? Here are some indicators.

1. Slowing Computer, Frequent Freezes, Blue Screen Of Death

This trio of PC malfunctions can have numerous causes, and a failing hard drive is one possibility. If these issues persist after a clean installation or in Windows Safe Mode, faulty hardware, particularly a failing hard drive, is likely to blame.

To rule out hard drive issues, you can utilize a range of diagnostic tools, but it's best to start by examining your system's S.M.A.R.T. (Self-Monitoring, Analysis, and Reporting Technology) data. To manually check your drive's S.M.A.R.T. status, you'll need a third-party tool like CrystalDiskInfo. Select the disk to